



Exploring the dynamics of toxigenic environmental *Vibrio mimicus* and its comparative analysis with *Vibrio cholerae* of the southern Gangetic delta

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Abstract

Vibrio mimicus and *Vibrio cholerae* are closely related species. Environmental *V. mimicus* were comparatively analyzed with *V. cholerae*, for the presence of virulence genes, antibiotic susceptibility, resistance genes, in-vitro hemolysis, and biofilm formation. Phylogenetic analysis was performed depending on toxin-gene disposition and isolation area. One *V. mimicus* isolate harbored *ctxA*, *tcp* El-Tor, *toxT* and *toxS*, whereas several strains contained incomplete copies of virulence cassettes and associated toxin genes. *V. cholerae* isolates harbored *ctx*, *tcp* and *toxT* genes, with a higher preponderance of *hlyA*, *rtxA* and *toxR* genes. *V. mimicus* were highly sensitive to amino/carboxy-penicillins, furazolidone & gentamycin, with quinolone & tetracycline resistance genes. *V. cholerae* isolates were sensitive to penicillins and cephalosporins, with 29% of the strains bearing the *sxt* gene. Phylogenetically, the apomorphic strains of both species were unique to the inland sites. *V. cholerae* has embodied an enormous public health burden globally but our findings emphasize the role of *V. mimicus* as an emerging etiological agent with similar epidemic potential.

Keywords Environmental *Vibrios* · Toxin genes · Antibiotic resistance · Hemolysis · PFGE analysis

Introduction

Vibrio mimicus has been a well-established pathogenic member of the genus *Vibrio*, responsible for various human illnesses (Chowdhury et al. 1989). Although credible evidence exists of a common ancestral evolutionary pathway of *V. mimicus* and *V. cholerae*, its genetic delineation from *V. cholerae* has been clearly documented (Hasan et al. 2010).

Both the species are inhabitants of freshwater as well as brackish aquatic environ and share some common pathogenic factors such as enterotoxins or hemolysins which leads to sporadic and sometimes epidemic diarrhoea and cholera (Shi et al. 1998; Chitov et al. 2009; Kay et al. 2012). Diarrhea, because of its higher incidence rate in both developed

and developing countries and significant epidemiological consequences, necessitates the surveillance of waterborne entero-pathogens (Mookerjee et al. 2014b; Halder et al. 2018). It has been demonstrated that under certain adverse environmental conditions, like the cells in a clustered biofilm or their colonization on chitinous substances, there is an increased chance of horizontal gene transfer among these pathogens (Halder et al. 2017; Batabyal et al. 2014b; Meibom et al. 2005). It has also been reported that *V. mimicus* could be the prime inheritors of gene transfer, thus making it the reservoir of virulence genes for other *Vibrio* species (Ramamurthy et al. 1994; Guardiola-Avila et al. 2015). Field et al. (2004) observed that the key to understanding the genetic diversity in an environment lies in the enumeration of the virulence factors of *V. mimicus* in its own entity. Alternatively, these virulence determinants of *V. mimicus* also need to be studied in conjunction with those of *V. cholerae*, because there have been evidences suggesting horizontal gene transfer aided by bacteriophages between the 2 *Vibrio* species (Boyd et al. 2000).

Pathogenic potential of any organism is established only when there is an expression of virulence associated genes, leading to the expression of the primary toxin

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genes (Meena et al. 2019), which makes the detection of associated factors equally important. Information regarding the presence of these virulence genes in *V.mimicus* of environmental origin, provides an insight toward the pathogenesis of this enteropathogen, and helps decipher the differences in its disease potential, compared with other *Vibrio*-borne gastroenteritis.

Furthermore, to the best of our knowledge, antibiotic resistance gene profile has not been investigated in *V.mimicus* species isolated exclusively from riverine sources, although there have been a few studies with antibiotic susceptibility reports on *V.mimicus*, isolated from seafood and human stool samples (Vaseeharan et al. 2005; Rebouças et al. 2011; Saha et al. 2020).

To decrypt the emergence of toxigenic *V.mimicus* and its relation, if any, to *V. cholerae* of the same environmental pool, the present study was undertaken to analyze bacterial isolates collected from the Gangetic south Bengal, where anthropogenic interventions and resultant contamination of river water are major perpetrators for making it a century old diarrhoea and cholera endemic foci (Mookerjee et al. 2014a; Batabyal et al. 2014a; Saha et al. 2019). From a diagnostic standpoint, pathogenic/non-pathogenic strains of *V. cholerae* can be differentiated by the presence of the *ctx* and *tcp* genes (Waldor and Mekalanos 1996). However, little is known about the toxigenic make-up of an environmental noncholera *Vibrio* strain like *V.mimicus*. Accordingly, our study aims to shed light on the distribution of the virulence and regulatory genes along with the antibiotic resistance gene profile of *V.mimicus*, which are more commonly associated with pathogenic *V.cholerae*, in an attempt to elucidate its horizontal gene transfer potential that may lead to the rise of a highly epidemic variant. Through this study we aim to understand the pathogenicity levels and disease potential of this emerging diarrhoeal etiological agent prior to formulating any imperative prevention measures.

Materials and methods

Study area and isolation

The *V.mimicus* and *V. cholerae* were isolated in a yearlong collection of environmental water samples from a flowing water milieu across 4 sites off the Gangetic delta, with distinct salinity gradients, namely Barrackpore, Howrah, Diamond Harbour on the bank of river Hooghly, and Gosaba along river Matla, West Bengal. Isolation and species identification were carried out by targeting the *dnaJ* gene of the *Vibrio* isolates, as per standardized protocol (Mookerjee et al. 2015; Nhung et al. 2007) (Tables 1 and 2). Subsequent assays were carried out in triplicates.

Biochemical characterization

The isolates were subjected to phenotypic characterization for initial species differentiation using these 7 tests: Voges Proskauer's, Arginine utilization, salt tolerance (1%), ONPG, citrate utilization, ornithine utilization & carbohydrate utilization (mannitol, arabinose, sucrose, glucose, salicin, cellobiose). The isolates were subjected to biochemical identification using commercially available miniaturized kits KB007 (HiMedia, Mumbai, India).

Polymerase chain reaction analysis of toxin genes

Polymerase chain reaction (PCR) was performed with purified genomic DNA as template extracted by the Boil-template method (Nandi et al. 2000), from all the isolated enteropathogenic *V.mimicus* and *V.cholerae* for the detection of the specific toxin genes associated with pathogenicity of the species. The reaction mixtures consisted of 10X PCR reaction buffer, 2.5 mM DNTP mixture, 1.5 mmol l⁻¹ MgCl₂, 100 nmol l⁻¹ of primer, 1U Taq polymerase (Merck Genie, Bangalore, India) and 3 µl of template DNA. Nucleotide sequence of the primers and the respective amplicon sizes are presented in Table 2.

Table 1 PCR based species-specific screening of *Vibrios*, targeting the *dnaJ* gene

Source	Riverine origin	Salinity gradient	<i>V.cholerae</i> n (%)	<i>V.mimicus</i> n (%)	Other ^a n (%)
Barrackpore	Hooghly	Low	27 (34.6)	13 (24.1)	09 (10.2)
Howrah	Hooghly	Low	16 (20.5)	27 (50.0)	14 (5.9)
Diamond Harbor	Hooghly	Medium	20 (25.6)	09 (16.6)	37 (42.0)
Gosaba	Matla	High	15 (19.2)	05 (9.2)	28 (31.8)
Total			78	54	88

^aOther includes *V. parahaemolyticus*, *V.alginolyticus* or *V. vulnificus*

Table 2 List of Primers

Target	Nucleotide sequence	Amplicon size	Reference
For the screening of the species-specific Vibrios targeting the <i>dnaJ</i> gene			
VM-Fa ^a	5'-CAGGTTTGYTGCACGGCGAAGA-3'		Nhung et al. 2007
<i>V.cholerae</i> ^b	5'-AGCAGCTTATGACCAATACGCC-3'	375 bp	
<i>V.mimicus</i> ^b	5'-YCTTGAAGAAGCGGTTTCGTGCA-3'	177 bp	
<i>V.parahaemolyticus</i> ^b	5'-TGCGAAGAAAGGCTCATCAGAG-3'	96 bp	
<i>V.alginolyticus</i> ^b	5'-GATCGAAGTRCCRACACTMGGA-3'	144 bp	
<i>V.vulnificus</i> ^b	5'-GTACGAAATTCTGACCGATCAA-3'	412 bp	
For the detection of Virulence & Regulatory Antibiotic Resistance Genes			
<i>vmh</i>	5'-GGTAGCCATCAGTCTTATCACG-3' 5'-ATCGTGTCCCAATACTTCACCG-3'	390 bp	Mookerjee et al. 2015
<i>ctxA</i>	5'-CTCAGACGGGATTTGTTAGGCACG-3' 5'-TCTATCTCTGTAGCCCCTATTACG-3'	302 bp	Mookerjee et al. 2015
<i>tcp El-Tor</i>	5'-GAAGAAGTTTGTAAGAAGAACAC-3' 5'-GAAAGGACCTTCTTTCACGTTG-3'	472 bp	Mookerjee et al. 2015
<i>tcpClassical</i>	5'-CACGATAAGAAAACCGGTCAAGAG-3' 5'-ACCAAATGCAACGCCGAATGGAGC-3'	617 bp	Mookerjee et al. 2015
NAG ST	5'-CCTATTTCATTAGCATAATG-3' 5'-CCAAAGCAAGCTGGATTGC-3'	215 bp	Ogawa et al. 1990
<i>tdh</i>	5'-GGTACTAAATGGCTGACATC-3' 5'-CCACTACCACTCTCATATGC-3'	199 bp	Mookerjee et al. 2015
<i>toxT</i>	5'-TTGCTTGGTTAGTTATGAGAT-3' 5'-TTGCAAACCCAGACTGATAT-3'	581 bp	Sechi et al. 2000
<i>toxR</i>	5'-GAAGCTGCTCATGACATC-3' 5'-AAGATCAGGGTGGTTATTC-3'	900 bp	Mookerjee et al. 2015
<i>toxS</i>	5'-CCACTGGCGGACAAAATAACC-3' 5'-AACAGTACCGTAGAACCGTGA-3'	640 bp	Sechi et al. 2000
<i>zot</i>	5'TCGCTTAACGATGGCGCGTTTT-3' 5'AACCCCGTTTCACTTCTACCCA-3'	947 bp	Rivera et al. 2001
RJ	5'TCGTTAGCGTGTTCGGTTCGCAGG 3'	1499	Mitra et al. 2001
LJ	5'TGCTTTGTACCAGTCACAGATAG 3' 5'GTGAATCTTGATGAGACGCTCTG 3' 5'GGTGAGCCAGGCTTATTTGGG 3'	1010	Mitra et al. 2001
<i>rtxA</i>	5'-CTG AAT ATG AGT GGG TGA CTT ACG-3' 5'GTG TAT TGT TCG ATA TCC GCT ACG-3'	417	Chow et al. 2001
<i>hlyA</i>	5'GGCAAACAGCGAAACAAATACC3' 5'CTCAGCGGGCTAATACGGTTTA3'	738	Singh et al. 2001
<i>tlc</i>	5' CTTTGGCCGTGTCTATTGGT3' 5' TAAATACGTGCGGCTCAACA3'		Bakhshi et al., 2012
<i>aldA</i>	5'ATTCTTCTGAGGATTGCTGAT3' 5'TTTTCTTGATTGTTAGGATGC3'	884	Boyd et al. 2000
For the detection of Antibiotic Resistance Genes			
<i>qnrVC</i>	5'-AATTTTAAGCGCTCAAACCTCCG-3' 5'-TCCTGTTGCCACGAGCATATTTT-3'	521 bp	Kim et al. 2010
SXT	5'-ATGGCGTTATCAGTTAGCTGGC -3' 5'- GCGAAGATCATGCATAGACC-3'	1035 bp	Okoh and Igbinsosa 2010
<i>sul2</i>	5'-AGGGGGCAGATGTGATCGC -3' 5'-TGTGCGGATGAAGTCAGCTCC -3'	625 bp	Okoh and Igbinsosa2010
<i>floR</i>	5'-TTATCTCCCTGTCTCCAGCG-3' 5'-CCTATGAGCACACGGGGAGC-3'	526 bp	Okoh and Igbinsosa 2010
<i>dfp18</i>	5'-TGGGTAAAGACACTCGTCATGGG-3' 5'-ACTGCCGTTTTCGATAATGTGG-3'	389 bp	Okoh and Igbinsosa 2010
<i>tetA</i>	5'-GTAATTCTGAGCACTGTCGC-3' 5'-CTGCCTGGACAACATTGCTT-3'	950 bp	Okoh and Igbinsosa 2010

Table 2 (continued)

Target	Nucleotide sequence	Amplicon size	Reference
<i>strB</i>	5'-GGCACCCATAAGCGTACGCC-3' 5'-TGCCGAGCACGGCGACTACC-3'	470 bp	Okoh and Igbinosa 2010
<i>dfrA1</i>	5'-CGAAGAATGGAGTTATCGGG-3' 5'-TGCTGGGGATTTCAGGAAAG-3'	372 bp	Okoh and Igbinosa 2010
For the detection of VPS genes			
<i>vpsR</i>	5'-TGGCGAAAGTGGTACTGGGAAAGA-3' 5'-CCAAGACCAAACAGCTCGCTTTCA-3'	911 bp	Gennari et al. 2012
<i>mbaA</i>	5'-GCATTCTAGAGTTCGCTACGATATGCTG-3' 5'-GCATGAATTCGCGTTGATAGGT GAATG-3'	516 bp	Bomchil et al. 2003

^aUniversal forward primer for 5 pathogenic *Vibrio* spp

^bReverse primers specific to the *Vibrio* species

Antibiotic sensitivity test

Antibiotic sensitivity was evaluated with all commonly prescribed antibiotics (namely ampicillin-10, ofloxacin-5, norfloxacin-10, ceftriaxone-30, amoxycillin-10, tetracycline-30, ciprofloxacin-5, nalidixic acid-30, kanamycin-30, meropenem-10, imipenem-10, azithromycin-15, gentamycin-10, cefuroxime-30, doxycycline-30, gatifloxacin-5, furazolidone-100, amoxycillin-clavulanic acid, cotrimoxazole-25, amikacin-30, carbenicillin-100, cefixime-5, chloramphenicol-30, cefotaxime-30) and expressed in mcg (Halder et al. 2018), applying the disk diffusion technique with commercially available antibiotic discs (BD Difco, Franklin Lakes, New Jersey, US) (Bauer et al. 1966). Results have been interpreted as per Clinical and Laboratory Standards Institute norms (Hindler and Richter 2016).

Analysis of antibiotic resistance gene

In an attempt to find a relationship between the multidrug-resistant phenotypes of *V. mimicus* and *V. cholerae*, along with the presence of the SXT-like element, PCR experiments were carried out using specific primers for the presence of antibiotic resistance genes (Table 2). To ascertain the contribution of SXT toward conferring of this resistance, we analyzed for the presence of *sul2* (encoding sulfamethoxazole resistance), *floR* (encoding chloramphenicol resistance), *dfr18* (O139 and non-O139 trimethoprim resistance), *strB* (aminoglycoside resistance), *tetA* (encoding tetracycline resistance), *qnrVC* (encoding quinolone resistance) and *dfrA1* (O1-specific trimethoprim resistance), the atypical clustered resistance genes, which are used to discriminate among SXT variants (Okoh and Igbinosa 2010).

Biofilm assay

To detect the biofilm production capability 2 different tests were carried out, namely a modified congo red assay and

another microtiter dish biofilm-formation assay. The isolates were inoculated onto Congo Red Agar plates, prepared using 5% w/v sucrose, Brain Heart Infusion Broth 3.7% w/v (Merck, Darmstadt, Germany), agar 1.5% and 0.1% congo red (indicator dye) (Sigma, St. Louis, US) (Halder et al. 2017). Plates were incubated at 30 °C for 24 h. The black color colonies with a dry crystalline morphology were considered positive for biofilm production, whereas the ones which remained pink with the absence of any dry crystalline form indicated negative for the biofilm formation (Odeyemi 2012). For the microtiter plate-based method, the protocol of GA O'Toole was followed, where biofilm formation via adherence is qualitatively measured using the crystal violet dye (O'Toole 2011). Those *V. mimicus* and *V. cholerae* isolates which were phenotypically positive for biofilm formation, were subjected to genotypic characterization using PCR (Table 2), for the presence of the *vspR* and *mbaA* genes, responsible for exopolysaccharide secretion and maintenance in *V. cholerae* (Gennari et al. 2012; Bomchil et al. 2003).

Hemolysis assay

To detect the functional expression of the hemolytic genes in these environmental *Vibrio* isolates in terms of their hemolytic activity, inoculation was carried out onto ready-to-use Sheep Blood agar plates (Himedia) and was incubated overnight (Halder et al. 2017). Those isolates which were devoid of any *hlyA*, *vmh*, and *tdh* genes were also inoculated as negative control and a standard *Staphylococcus aureus* strain (ATCC 25,923) was used as positive control.

Pulsed field gel electrophoresis assay

Phylogenetic relatedness of the toxin-gene bearing *V. mimicus* and *V. cholerae* isolates, using PFGE analysis, has been deduced as per the CDC PulseNet standardized protocol for Vibrios (Warning 2017). Briefly, colonies of

each strain were suspended in 2 mL of Tris EDTA buffer (100 mM, pH 8.0) and adjusted to 0.8 absorbance at 610 nm. 400 μ L of the adjusted bacterial cell suspension was mixed with 20 μ L (20 mg/mL) of proteinase K and 400 μ L of 1% molten agarose (Seakem Gold; Lonza Bioscience, Pennsylvania, US) and poured into plug molds for solidification. Plugs were subsequently submerged in a lysis buffer (50 mM Tris, 50 mM EDTA, pH 8.0, 1% Sarcosyl), and incubated in a shaking water bath for 1 h at 54 °C, followed by washing in 6 phases, twice with sterile water and 4 times with 1X TE buffer. Subsequent restriction digestion was carried out with *NotI* and *XbaI* (40U/ μ L each) (Takara Bio, Shiga, Japan) for *V.mimicus* isolates and *Salmonella ser. braenderup* H9812 (used as DNA marker), respectively, and incubated at 37 °C. Plugs were run on 1% agarose gel in 0.5X TBE buffer in a PFGE equipment (CHEF DR-III; Bio-Rad, Hercules, California, US), pulse time ranged between 2–10 s for 13 h and 20–25 s for 6 h. PFGE banding patterns of the *Vibrio* isolates were examined applying BioNumerics software (version 6.1; Applied Maths, Kortrijk, Belgium), and a dendrogram was constructed using the Dice similarity coefficient and the unweighted pair-group method with arithmetic mean algorithm (UPGMA dendrogram) with a position tolerance of 1.3%.

Results

Biochemical characterization

All the *V.mimicus* isolates collected from the riverine water samples showed positive reactions for salt tolerance (at 1%), ONPG, citrate and ornithine utilization while showing negative utilization reactions for Voges Proskauer (meaning no acetoin production ability) and arginine decarboxylation.

Amongst the carbohydrate reactions, barring mannitol and glucose, the isolates showed negative utilization reactions to all other carbohydrate substrates that they were subjected to, namely sucrose, arabinose, salicin, and cellobiose.

All of the *V. cholerae* isolates showed positive Voges Proskauer reaction along with 100% ornithine, mannitol and glucose utilization. The isolates showed salt tolerance at 1% NaCl. However, galactose, inulin, sodium gluconate, glycerol, salicin, esculin, citrate could be utilized by < 50% of the isolates. Unlike the *V.mimicus* isolates, all the *V.cholerae* showed positive utilization reaction for sucrose.

PCR analysis

A total of 54 *V.mimicus* and 78 *V.cholerae* isolates collected from 4 different sites on the Hooghly and Matla river were studied for the presence of virulence genes, sensitivity to various antibiotics, presence of antibiotic resistance genes,

along with their capacity for the formation of biofilm, and their hemolytic abilities. The site-wise distribution of virulence and regulatory genes is shown in Fig. 1.

Antibiotic sensitivity test

The antibiotic susceptibility test was performed for all the environmental *V.mimicus* and *V.cholerae* isolates against all standard antibiotic dosages. Notably, despite the CLSI guidelines stating that due to the limited availability of clinical data for doxycycline and azithromycin, the utility of the interpretive criteria for *Vibrio* spp. other than *V. cholerae* remains uncertain (Hindler and Richter 2016), doxycycline and azithromycin were incorporated in the assay panel while carrying out the antibiogram tests. This is because of the fact that, widespread administration of a combination of third-generation cephalosporin and tetracycline or one of its analogs like doxycycline, in cases of infections caused by noncholera *Vibrios* (other than gastroenteritis), still remains the therapy of choice in adult patients (Ho, *Vibrio* Infections Medication 2021). Furthermore, those clinical *V. cholerae* isolates susceptible to tetracycline are usually susceptible to doxycycline (Hindler and Richter 2016) and single dose azithromycin has been proved to be effective in the treatment of cholera in children, which made it imperative that we test these environmental isolates for their sensitivity to doxycycline/azithromycin.

The sensitivity patterns of *V.mimicus* are presented in Fig. 2 A. Irrespective of the retention of toxin genes the isolates were found to be sensitive to ampicillin, amoxycillin, meropenem, imipenem, azithromycin, gentamycin, furazolidone and carbenicillin. The isolates were moderately sensitive to gatifloxacin and chloramphenicol, with a resistance toward some of the fluoroquinolones, tetracyclines and cephalosporins (upto 80% in case of ofloxacin, norfloxacin, doxycycline, and cefuroxime).

The antibiotic susceptibility patterns of *V.cholerae* exhibited high resistance ($\geq 60\%$) to fluoroquinolones, quinolones, cotrimoxazole, chloramphenicol and amoxycillin-clavulanic acid. Notably, 100% of the *V.cholerae* isolates were found to be resistant against tetracyclines. However, most *V. cholerae* isolates were sensitive to penicillin and cephalosporin derivatives (Fig. 2 B).

Antibiotic resistance genes

Amongst the 54 *V.mimicus* isolates, 43 isolates showed resistance to the quinolone, tetracycline, and cephalosporin groups, of which 29 isolates were multi-drug resistant (MDR), that is resistant to 2 or more groups. Among the MDR isolates, 2 isolates were found to have both *qnrVC* (responsible for quinolone resistance) and *tetA* (responsible for tetracycline resistance). Surprisingly, amongst those

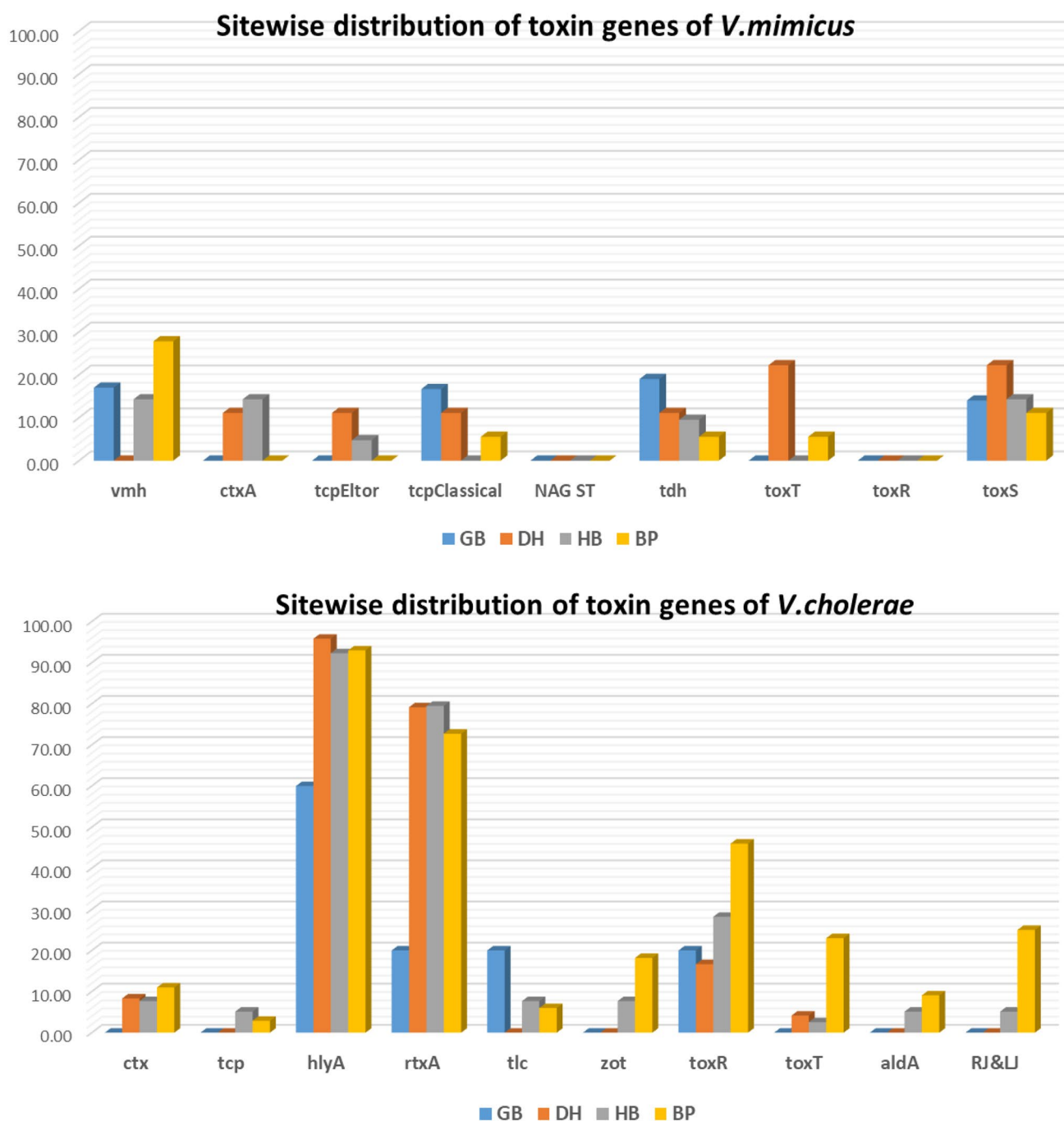


Fig. 1 Site-wise distribution of virulence and regulatory genes in the environmental *V. mimicus* and *V. cholerae* isolates

isolates which showed resistance to only 1 group of antibiotics, 3 isolates showed resistance to *tetA*. On the contrary, 23 out of the 78 *V. cholerae* isolates were detected as

sxt positive and 8 isolates harbored additional resistance genes, viz. *tetA* (4), *sul2* (2) and *qnrVC* (5).

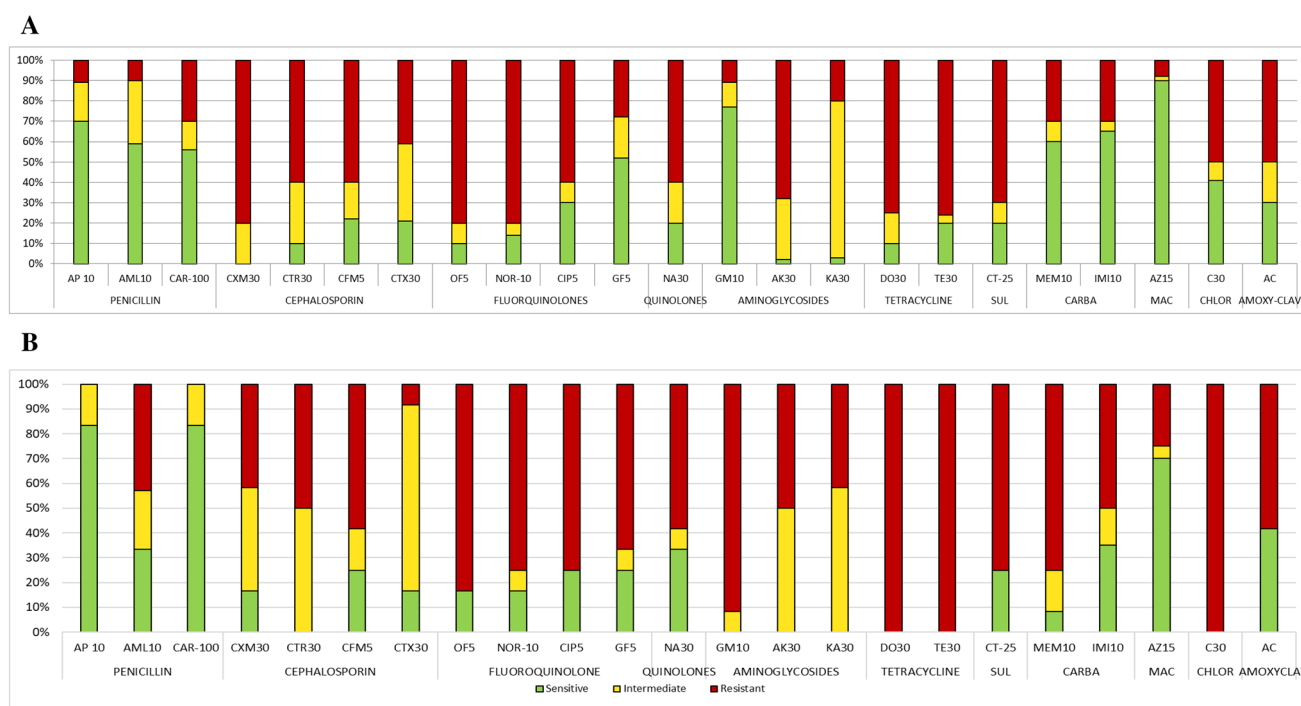


Fig. 2 Antibiotic sensitivity pattern of the environmental *Vibrio* isolates. (AP ampicillin, OF ofloxacin, NOR norfloxacin, CRO ceftriaxone, AML amoxicillin, TET Tetracycline, CIP ciprofloxacin, NA nalidixic acid, KA kanamycin, MR meropenem, IMI imipenem, AZ

azithromycin, GM gentamycin, CXM cefuroxime, DO doxycycline, GF gatifloxacin, FZ furazolidone, AC amoxycillin-clavulanic acid, CT cotrimoxazole, AK amikacin, CB carbenicillin, CFM cefixime, C chloramphenicol, CX cefotaxime)

Biofilm assay

Only those isolates, which were positive for both the assays, were taken into consideration. Out of the 54 *V. mimicus* isolates and 78 *V. cholerae* isolates, 17 (31.48%) and 19 (24.35%) were positive for biofilm assay respectively, that is, they have the potential for producing biofilm. Interestingly, most of the environmental isolates that were found to produce the biofilm were found to be positive for the presence of some of the virulence as well as regulatory genes, viz. *ctxA*, *tdh*, *toxR* and *toxS*. Only 3 *V. mimicus* and 2 *V. cholerae* isolates harbored the *vspR* gene whereas the *mbaA* gene could be detected in 1 *V. mimicus* following PCR analyses of the biofilm positive isolates.

Hemolysis test

All 54 *V. mimicus* were tested for hemolysis activity on Sheep Blood agar plates. Amongst them, 9 of the 14, that were PCR positive for the hemolysin gene (either 1 or both *vmh* and *tdh*), could break down RBCs and subsequently exhibit a lysis zone. Surprisingly, 3 amongst the other 40 isolates which were PCR negative for the hemolysin genes also showed hemolytic activity on the blood agar plates.

Amongst the *V. cholerae* isolates, 24 (of 78) showed hemolytic activity, of which only 18 were found to harbor hemolytic genes (*hlyA*). Interestingly however, majority of the *hlyA* positive *V. cholerae* isolates did not express hemolysis on blood agar plates.

PFGE analysis

Dendrograms were constructed from PFGE profiles of both *V. mimicus* and *V. cholerae* (Fig. 3), clustered as genotypes with similarity over 85%, and the isolates were assigned numbers from VM1 to VM20 and VC1 to VC20 for *V. mimicus* and *V. cholerae*, respectively.

A total of 12 different profiles were obtained for *V. mimicus*, clustered as 9 different genotypes. The dendrogram analysis revealed a distinct but closely related genotype based on the point of isolation; those isolated from low-saline zones like Howrah and Barrackpore are more closely related than those obtained from high saline zone like Gosaba.

Most of the high saline *V. cholerae* isolates from Gosaba were clustered separately from the fresh water inland sites, viz. Howrah and Barrackpore. All of the toxigenic isolates from Diamond Harbour, Howrah and Barrackpore sites were closely related, of which Howrah and

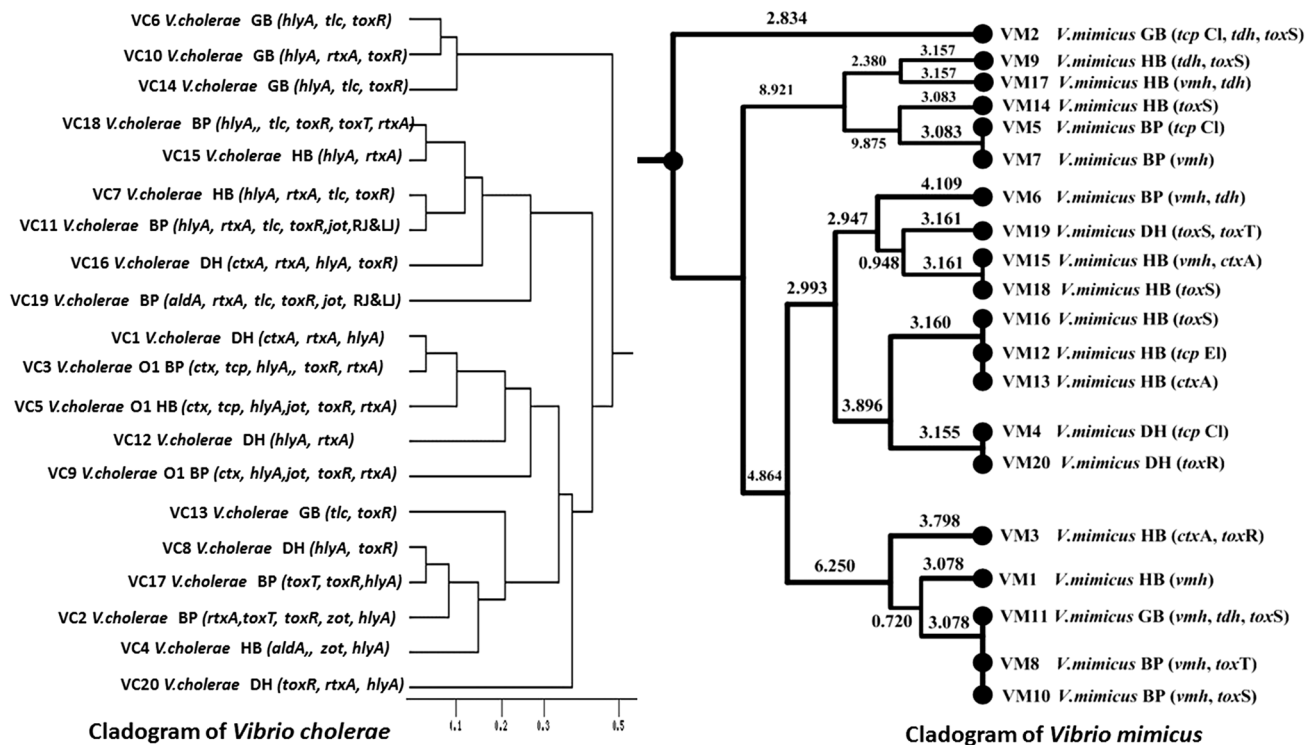


Fig. 3 Dendrogram depicting the genetic diversity among various environmental toxigenic *V. mimicus* and *V. cholerae* strains, constructed by the unweighted pair-group method using averages. (VM1-

20 *V.mimicus* isolates, VC1-20 *V.cholerae* isolates, HB howrah, DH diamond Harbor, BP barrackpore, GB gosaba)

Barrackpore sites harbored highly toxigenic *V. cholerae* isolates and shared high genetic similarities amongst them.

Discussion

Our exploration of toxin genes from *V. mimicus* of environmental origin revealed that there is presence of *ctxA* and *tcp* genes (both *El-Tor* and *Classical*) in some of the *V. mimicus* isolates, which increases the chances of diarrheal episodes manifold. These have been repeatedly implicated to be the causative toxins of epidemic cholera in *V. cholerae* O1/O139 (Aliabad et al. 2012). Further analysis of environmental isolates showed a comparatively higher distribution of *toxS*, *vmh* and *tdh* and a rarer distribution of *toxT* genes, and the absence of any *toxR* or NAG ST genes. A comparative analysis of the toxin genes found in the *V. cholerae* isolates of the same aquatic niche showed the presence of *ctx*, *tcp*, and *toxT* genes in a small number of isolates, with a higher preponderance of *hlyA*, *rtxA* and *toxR* genes.

This definitive but narrow dissemination of the *ctx*, *toxT* and *tcp* genes among environmental *V. mimicus* isolated from the same aquatic pool as that of the toxin-gene bearing *V. cholerae*, may have been the result of an incomplete horizontal transfer of genes between the 2 *Vibrio* species.

ToxR and *toxS* genes are involved in the co-regulation of a number of factors in addition to *tcp* and *ctx* (DiRita et al. 1991). Transcription of the operon encoding these genes is responsive to environmental signals such as temperature, pH etc., which also affects gene acquisition in the environmental pool of *Vibrios* in a catastrophic environment (Palit and Batabyal 2010; Guijarro et al. 2015; Mookerjee et al. 2020). This suggests a distinct possibility of a similar phenomenon occurring in the aquatic environ at optimum levels of temperature and pH, which might also activate the toxin genes in *V. mimicus* isolates. This singularity converts the environmental *V. mimicus* pool into a reservoir of toxin genes (*ctx*) in the riverine system and it confers proficiency to transfer these genetic elements to non-pathogenic strains of *Vibrios*, rendering them pathogenic. Anthropogenic behavioral practices result in a situation of close proximity between community and the organisms, resulting in its ingestion, occurrence of diarrhoeal disease, and resultant morbidity.

To the best of our knowledge, pathogenic *Vibrios* of clinical origin and those isolated from seafood, have been well studied and documented; however our study comprised the isolates of riverine-estuarine origin, which are yet to be comprehensively explored. Here, detection of *tdh* gene amongst environmental *V. mimicus* isolates was demonstrated, which collaterally proves that these isolates harbor

the direct hemolysin gene and effectively carries the possibility that *V. mimicus* may be a potent reservoir of *tdh* gene in the environment. Thus far, *tdh* gene in *V. mimicus* has been reported only from clinical isolates (Shinoda et al. 2004).

Although only 4 *V. mimicus* and 2 *V. cholerae* nonO1 isolates showed the presence of *vspR* and *mbaA* gene, about one-third of the isolates, including some of those harboring toxin genes, were phenotypically positive for biofilm formation. The reason may be attributed to the presence of a myriad of other toxin genes, specific to these environmental noncholera *Vibrios*, responding to various eco-hydrological signals, such as temperature and pH. There have been recent studies on *S. aureus*, demonstrating that hemolysin and some other virulence factors, apart from specific biofilm-formation genes, participate in the formation of biofilm (Den Reijer et al. 2016). The biofilm acts as a substratum which might aid in the horizontal transfer of these genes between the toxic and non-toxic *V. mimicus* isolates as tested and documented in the case of *V. cholerae* when confined to such a community (Meibom et al. 2005; Blokesch and Schoolnik 2007). It has been shown that biofilms contribute to enhanced infectivity and environmental resilience of pathogenic *Vibrio* species (Meibom et al. 2005). Most of the biofilm positive isolates of both *V. mimicus* and *V. cholerae*, belonging to our study, were also found to harbor some of the important virulence regulatory genes like *tdh*, *toxR*, and *toxS*. These 2 factors in conjugation may act as a driving force, enhancing the waterborne spread of bacterial epidemics, and aiding in the sustainance of the epidemic strains in the aquatic milieu of this study foci of deltaic West Bengal.

One well-acknowledged virulence marker that has always been epidemiologically associated with disease is the beta type hemolysis produced on Sheep Blood agar plates. Almost all *tdh* positive *V. mimicus* strains isolated from clinical specimens demonstrate this hemolytic activity, whereas only about 1% to 2% of strains from non-clinical sources show hemolysis (Nishibuchi et al. 1991; Shinoda et al. 2004). Here we report that 9 (64%) out of 14 hemolysin gene bearing environmental *V. mimicus* isolates showed beta hemolysis on Sheep Blood agar plates. The intermediate or negative hemolysis in the 5 *tdh/vmh* bearing isolates may be explained by the low or no expression of either of the 2 hemolysin genes, or the strains being non- or low- virulent (Nishibuchi and Kaper 1995; León Robles et al. 2013). The 9 isolates showing β hemolysis, signifies a high-level of TDH/VMH production leading to a positive phenotypic expression on the blood agar plates. Although unexpected, 3 of the strains which did not harbor either the *tdh* or the *vmh* gene, showed clear beta hemolysis, which may be due to the expression of some other distinct hemolysin gene that *V. mimicus* has been shown to possess (Hasan et al. 2010). Interestingly, only about 45% (24/53) of the *hlyA* gene bearing *V. cholerae* isolates showed distinct hemolysis on blood

agar plates, affirming that not all the *hlyA* positive isolates possess hemolytic activity. Nonetheless, the fact remains that the presence of enterotoxigenic/cytotoxic *V. cholerae* isolates in a certain aquatic reservoir, which also harbors 'genetic inheritors' like that of *V. mimicus*, always amplifies the chances of inheritance and transmission of virulence.

Most of the isolates were highly sensitive to the extended-spectrum aminopenicillins, which have normally been conferred resistant in most parts of the world (Center and for Disease Dynamics, Economics Policy, CDDEP 2018). This also suggests that these isolates have not yet had any gut exposure, and nor were they exposed to the antibiotics in the aquatic environment. Previous studies have shown that tetracyclins and aminoglycosides were active against environmental *Vibrio* species (Okoh and Igbinosa 2010), but this was in contrast with our observation in *V. mimicus* isolates where we detected resistance to tetracycline, doxycycline, and amikacin. We detected that about 70% of the isolates were resistant to the standard extensively-prescribed drug combinations of choice like amoxycillin-clavulanic acid, doxycycline, and broad-spectrum cephalosporins like cefuroxime, which is a matter of serious concern given the invasion opportunity from its niche to the community during the course of regular contact.

Following rehydration and nutritional supplementation, antibiotics remain the primary treatment stratagem to combat cholera or diarrhoea, following resolution of the initial fluid deficit; but the global health issue of antimicrobial drug resistance (multi-drug resistance or MDR in case of resistance to 2 or more antimicrobials) constantly challenges this approach. 79% (43/54) of the *V. mimicus* isolates in our study showed resistance to 1 or more of these 3 groups of antibiotics, namely, quinolone, tetracycline and cephalosporin groups, of which 53% (29/54) were MDR. 2 of these MDR *V. mimicus* isolates were shown to harbor the genes conferring resistance to quinolone and tetracycline group, while 3 of the isolates showing resistance to only 1 group of antibiotics, harbored the tetracycline resistance gene. 29% and 10% of the *V. cholerae* isolates in our study harbored *sxt* gene and associated antibiotic resistance genes (*tetA*, *sul2*, and *qnrVC*), respectively. Although SXT has been found in most clinical and environmental isolates of *V. cholerae*, some of these genes were solely described in clinical isolates (Kim et al. 2010). But the unique detection of these antimicrobial resistance genes from environmental *V. mimicus* of a flowing water system, possibly explains dissemination of these genes among environmental *Vibrios* and other enteropathogenic bacteria present in the same milieu.

The total number of different PFGE profiles found in the environmental isolates of *V. mimicus* was significantly higher than those identified in clinical isolates in similar studies elsewhere, providing evidence of co-existence of multiple genotypes. The isolates showed distinct

segregation with the zone of isolation with a greater number of related toxin-gene bearing isolates being obtained from low-saline zones as compared to high saline study zones like Gosaba. In the case of *V.cholerae*, the dendrogram revealed a clear distinction between the strains based on the salinity gradient of the point of isolation. Furthermore, the low-saline toxigenic strains were more closely related as compared to those isolated from Gosaba. From the dendrogram obtained, it becomes clear that the more primitive strains or plesiomorphic *Vibrio* strains have originated from Gosaba, making them the 'outgroup,' while the apomorphic strains are unique to the inland sites (Barrackpore and Howrah) where there is more anthropogenic intrusion, industrial influence, and a plethora of other bacterial entities, making them more susceptible to acquiring innumerable genetic traits.

Conclusion

As befitting of a pathogen, capable of existing freely without the support of a host in an ever-changing aquatic milieu, *V.mimicus* is an emerging threat, to the community. On the basis of our findings, we propose that given suitable environmental triggers, as demonstrated earlier (Batabyal et al. 2016), it can spill over into human populations as an incipient etiological agent leading to explosive diarrhoeal outbreaks, of which, till date, *V.cholerae* has been the primarily implicated organism. The presence of toxigenic *V.cholerae* isolates in the aquatic environment along with this morbidic *V.mimicus* pool, establishes the fact that diarrheal disease epidemiology is largely linked to environmental reservoir of dormant yet potentially pathogenic pool of these *Vibrios*. As is the case with most of the third world countries, active disease monitoring and surveillance is undertaken only as a consequence or the upsurge of a certain disease. This is primarily due to limited finance, poor health policies, and an ever-increasing population. But because cholera was a constant occurrence in these parts of the globe, with increased vigilance over the years and better medical interventions, cholera mortality has reduced over the years; however, the morbidity rate still remains high, with India being among the top 5 vulnerable countries. What is of more pressing concern is whether the void created by the reduction in cholera, is being taken up by another silently emerging pathogen, i.e., *V.mimicus*. The only point of solace for clinicians is that these organisms are still sensitive to most of the conventional antibacterial drugs. In the current scenario, rational use of antibiotics, prohibition of unregulated, over-the-counter access to antimicrobials, and relentless monitoring must be undertaken. On the contrary, this also leads to an imperative situation where dissemination of the toxin genes in the

flowing aquatic system need to be explored unceasingly, to understand the pathogenicity levels and disease potential, prior to drawing any obligatory preventive measures against the diarrheal etiological agents of bacterial origin, like *V.mimicus* and *V.cholerae*.

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